

chrF score

Q. What is corpus-level chrF ?

→ suppose we have three sentences

Reference SK sentences	Hypothesis SK sentences
1. 아내가 학교에 도착했다.	안해가 학교에 도착했다.
2. 내일 선생님을 만날 예정이다.	내일 선생님을 만날 예정이다.
3. 초밥을 켜다.	초밥을 켜다.

$$\frac{\text{chrF}(\text{sentence 1}) + \text{chrF}(\text{sentence 2}) + \text{chrF}(\text{sentence 3})}{3}$$

The general formula for the CHRF score is:

$$\text{CHRF}\beta = (1 + \beta^2) \frac{\text{CHRP} \cdot \text{CHRR}}{\beta^2 \cdot \text{CHRP} + \text{CHRR}}$$

"Corpus counts → Corpus precision → Corpus recall → chrF"

Character n-gram Precision (CHRP)

For each character n-gram order (e.g., 1-grams, 2-grams, ..., 6-grams), precision is computed as

$$P_n = \frac{\text{number of matching character n-grams}}{\text{number of character n-grams in the hypothesis}}$$

Measures how many character n-grams generated by the model (hypothesis) also appear in the reference at that particular n-gram order.

The final character precision is obtained by averaging across all n-gram orders:

$$\text{CHRP} = \frac{1}{N} \sum_{n=1}^N P_n$$

When computing P_n for $n = 1 \dots 6$, consider all sentences in the corpus

e.g., P_1

- sentence 1	- sentence 2	- sentence 3
matches = 9	matches = 13	matches = 5
hyp chars = 11	hyp chars = 13	hyp chars = 6

the counts are added all together !

$$\begin{aligned} \text{total matches} &= 27 & P_1 &= 27/30 = 9/10 = 0.9 \\ \text{total hyp chars} &= 30 \end{aligned}$$

repeat this process for $P_2, \dots, P_6$

$$\text{CHRP} = (P_1 + \dots + P_6) / 6$$

Character n-gram Recall (CHRR)

Similarly, recall is computed separately for each character n-gram order:

$$R_n = \frac{\text{number of matching character n-grams}}{\text{number of character n-grams in the reference}}$$

Measures how many character n-grams in the reference are successfully captured by the hypothesis at that particular n-gram order.

The final character recall is obtained by averaging across all n-gram orders:

$$\text{CHRR} = \frac{1}{N} \sum_{n=1}^N R_n$$

When computing R_n for $n = 1 \dots 6$, consider all sentences in the corpus

e.g., R_1

- sentence 1	- sentence 2	- sentence 3
matches = 9	matches = 13	matches = 5
ref chars = 11	ref chars = 13	ref chars = 6

the counts are added all together!

total matches = 27

$$R_1 = 27 / 30 = 9 / 10 = 0.9$$

total ref chars = 30

repeat this process for $R_2, \dots, R_6$

$$\text{CHRR} = (R_1 + \dots + R_6) / 6$$

$\therefore$ end up with one chrF score for the corpus



If we want to include whitespaces as characters, use `whitespace = True`

```
from sacrebleu.metrics import CHRF
chrf = CHRF(word_order=0, beta=3)
print(chrf.sentence_score("책상 밑으로", ["책상 밑으로"]).score)
✓ 0.0s
100.0
```

VS.

```
from sacrebleu.metrics import CHRF
chrf = CHRF(word_order=0, beta=3, whitespace=True)
print(chrf.sentence_score("책상 밑으로", ["책상 밑으로"]).score)
✓ 0.0s
34.32572050027189
```

! chrf score is penalized a lot because "책상 밑으로" is a short sequence

Reference SK sentences	Hypothesis SK sentences
책상 밑으로	책상밑으로

• Unigrams

$$P_1 = 5/5 = 1$$

$$R_1 = 5/6 \approx 0.83$$

• Trigrams

hyp	ref
책상밑	책상_
상밑으	상_밑
밑으로	_밑으
	밑으로

$$P_3 = 1/3 \approx 0.33$$

$$R_3 = 1/4 = 0.25$$

• 4-grams

hyp	ref
책상밑으	책상_밑
상밑으로	상_밑으
	_밑으로

$$P_4 = R_4 = 0$$

• Bigrams

hyp	ref
책상	책상
상밑	상_
밑으	_밑
으로	밑으
	으로

$$P_2 = 3/4 = 0.75$$

$$R_2 = 3/5 = 0.6$$

• 5-grams

hyp	ref
책상밑으로	책상_밑으
	상_밑으로

$$P_5 = R_5 = 0$$

c.f. no 6-grams because the hypothesis has only 5 characters

$$CHRP = \frac{1 + 0.75 + 0.33 + 0 + 0}{5} \approx 0.416$$

$$CHRR = \frac{0.83 + 0.6 + 0.25 + 0 + 0}{5} \approx 0.336$$

$$CHRF_3 = 10 \times \frac{0.416 \times 0.336}{9 \times 0.416 + 0.336} \approx \frac{1.397}{4.08} \approx 0.34$$

∴ chrf is highly sensitive to spacing differences in short sequences because a single space disrupts a large proportion of their higher-order character n-grams

in the above example, a single spacing difference destroyed most of the phrase's available 4-gram and 5-gram overlap!

❖ chrF++

evaluates MT output based on character n-gram precision and recall enhanced with word n-grams

* default

character n-grams : 6

- CHRP = averages character unigram precision, bigram precision ... 6-gram precision
- CHRR = averages character unigram recall, bigram recall ... 6-gram recall

word n-grams : 2

- WORDP = averages word unigram precision and word bigram precision
- WORDR = averages word unigram recall and word bigram recall

$$P = \frac{CHRP + WORDP}{2}, \quad R = \frac{CHRR + WORDR}{2}$$

$$\text{chrF++} = (1 + B^2) \frac{P \cdot R}{B^2 P + R}$$